

# MyHomeUSB CRM — API & Data Model Reference

\_\_Backend surface area (FastAPI) + MongoDB schema.\_\_

## 1. Routers Overview

| Router             | Path prefix                                     | Purpose                                                                      |
|--------------------|-------------------------------------------------|------------------------------------------------------------------------------|
| `auth.py`          | `/api/auth`                                     | Login, register, JWT issue, password reset                                   |
| `crm.py`           | `/api/crm`                                      | Leads, pipeline, distribution rules                                          |
| `projects.py`      | `/api/projects`                                 | Master projects, WOs, stages, additional work, RAB lifecycle (12,000+ lines) |
| `financial.py`     | `/api/accountant` + `/api/cre`                  | Cashbook, approvals, cheques, income, suspense                               |
| `operations.py`    | `/api/accountant/cheques` etc.                  | Cheque Management 7 tabs, vendor book                                        |
| `site_ops.py`      | `/api/site-engineer`, `/api/procurement-simple` | SE board API, material/labour requests, petty cash                           |
| `procurement.py`   | `/api/procurement`                              | Procurement-side dashboards, vendor book                                     |
| `client_portal.py` | `/api/client-portal`                            | Client-only API                                                              |
| `public.py`        | `/api/public`                                   | Public lead form, package gallery                                            |
| `hr.py`            | `/api/hr`                                       | Attendance, leave, payroll (Phase 2)                                         |

All routes are JWT-authenticated except `/api/auth/login` and `/api/public/*`.

## 2. Key Endpoints by Module

### 2.1 Authentication

- `POST /api/auth/login` — body: `{email, password}` → `{token, user}`.
- `POST /api/auth/register` — admin-only.
- `GET /api/auth/me` — current user profile.

### 2.2 Projects (subset)

- `GET /api/projects` — list (role-scoped).
- `POST /api/projects` — create (CRE).
- `GET /api/projects/{id}` — detail.
- `PATCH /api/projects/{id}/status` — advance status (with role gating).
- `GET /api/projects/{id}/work-orders` — list WOs.
- `POST /api/projects/{id}/work-orders` — create WO.
- `GET /api/projects/{id}/payment-stages` — client-side payment ladder.

### 2.3 Work Order Scope (Feb 2026)

- `POST /api/projects/{pid}/work-orders/{wid}/scope-items` — add scope line.
- `DELETE /api/projects/{pid}/work-orders/{wid}/scope-items/{idx}` — delete line.

- `POST /api/projects/{pid}/work-orders/{wid}/stages` — add stage (fixed / percentage).
- `DELETE /api/projects/{pid}/work-orders/{wid}/stages/{sid}` — delete stage (refuses if RABs exist).

## 2.4 RAB lifecycle

- `POST /api/projects/{pid}/work-orders/{wid}/stages/{sid}/payment-requests` — raise RAB.
- `POST /api/projects/{pid}/work-orders/{wid}/multi-stage-payment-requests` — multi-stage RAB.
- `PATCH /api/projects/{pid}/work-orders/{wid}/stages/{sid}/payment-requests/{rid}` — edit pending RAB (amount, notes, target\_stage\_id).
- `POST /api/projects/{pid}/work-orders/{wid}/stages/{sid}/payment-requests/{rid}/pm-approve` — PM step.
- `.../qc-approve` — QC step.
- `.../planning-approve` — Planning step.
- `POST /api/accountant/labour-payments/{rid}/release` — Accountant release. Body supports `sibling_request_ids[]` for multi-stage atomic release + `payment_entries[]` for multi-mode.
- `GET /api/accountant/labour-rab/{rid}/pay-context` — pre-release context (siblings, suspense balance).

## 2.5 Additional Work

- `POST /api/projects/{pid}/work-orders/{wid}/additional/sections` — create section.
- `DELETE /api/projects/{pid}/work-orders/{wid}/additional/sections/{sid}` — delete section.
- `PATCH /api/projects/{pid}/work-orders/{wid}/additional-sections/{sid}/lock` — lock/unlock section (cascades to items + section stage).
- `POST /api/projects/{pid}/work-orders/{wid}/additional` — add item.
- `PATCH /api/projects/{pid}/work-orders/{wid}/additional/{idx}` — edit item.
- `DELETE /api/projects/{pid}/work-orders/{wid}/additional/{idx}` — delete item.
- `PATCH /api/projects/{pid}/work-orders/{wid}/additional/{idx}/lock` — per-item lock (gated on section being unlocked).

## 2.6 Cheques

- `GET /api/accountant/cheques?status=...` — list (with project\_name backfill).
- `POST /api/cre/cheques` — CRE submits.
- `POST /api/accountant/cheques/{id}/open` — Accountant opens.
- `PATCH /api/accountant/cheques/{id}/issue` — issue to vendor.
- `PATCH /api/accountant/cheques/{id}/bounce` — mark bounced.

## 2.7 Site Engineer

- `GET /api/site-engineer/project/{pid}` — SE/PM/Super Admin view of a project.
- `POST /api/site-engineer/material-requests`.
- `POST /api/site-engineer/labour-payments`.
- `POST /api/site-engineer/petty-cash`.

## 2.8 Client Portal

- `GET /api/client-portal/my-projects`.

- GET /api/client-portal/projects/{id}/dashboard.
- POST /api/client-portal/additional-costs/{cid}/approve //reject.

### 3. MongoDB Collections (selected fields)

`users`

```
{
  "user_id": "usr_abc",
  "email": "anbu@my.com",
  "name": "Anbu",
  "role": "site_engineer",
  "phone": "...",
  "is_active": true,
  "password_hash": "bcrypt$2b$..",
  "slot_id": "slot_morning",
  "created_at": "ISO",
  "metadata": {...}
}
```

`projects`

```
{
  "project_id": "proj_xxx",
  "project_number": "P-2026-001",
  "name": "anbu",
  "client_id": "client_abc",
  "client_name": "Anbu",
  "location": "...",
  "latitude": 11.1, "longitude": 77.0,
  "status": "active",
  "current_stage": "sub_structure",
  "team": [{"user_id": "...", "role": "...", "name": "...}],
  "contract_value": 5000000,
  "created_at": "..."
}
```

`project\_work\_orders`

```
{
  "work_order_id": "wo_xxx",
  "project_id": "proj_xxx",
  "contractor_id": "cont_abc",
  "contractor_name": "Alli muthu",
  "contractor_type": "Civil",
  "status": "active",
  "scope_items": [{"name": "...", "unit": "sqft", "quantity": 100, "unit_rate": 50, "total": 5000}],
  "scope_total": 5000,
  "additional_sections": [{"section_id": "sec_xxx", "name": "anbu Demo Section", "claim_type": "rework_se", "is_locked": false}],
  "additional_work": [{"description": "...", "unit": "nos", "quantity": 1, "unit_rate": 500, "total": 500, "section_id": "sec_xxx", "c...}],
  "additional_total": 500,
  "deduction_total": 0,
  "total_value": 5500,
  "stages": [{
    "stage_id": "stg_xxx",
    "stage_label": "S1",
    "name": "Foundation Completion",
    "amount": 1000,
    "scheduled_amount": 1000,
    "is_open": true,
    "is_addition": false,
    "claim_type": null,
    "linked_section_id": null,
    "payment_requests": [{
      "request_id": "req_xxx",
      "rab_number": "RAB-01",
      "rab_group_id": "grp_xxx",
      "amount": 1000,
      "approved_amount": 1000,
      "status": "approved",
      "is_multi_stage": false,

```

```

    "stage_breakdown": [],
    "pm_approved_by_name": "...",
    "qc_approved_by_name": "...",
    "planning_approved_by_name": "...",
    "payment_record": {...},
    "notes": "...",
    "timeline": [{"step": "requested", "at": "..."}]
  }],
  "amount_released": 1000,
  "amount_pending": 0
}],
"paid_amount": 1000
}

```

## `cheques`

```

{
  "cheque_id": "chq_xxx",
  "cheque_no": "123456",
  "amount": 50000,
  "cheque_date": "2026-01-15",
  "bank": "HDFC",
  "drawer_name": "...",
  "project_id": "proj_xxx",
  "project_name": "anbu",
  "status": "received | opened | awaiting_cre | issued | bounced | disabled",
  "purpose": "income | issued_payment",
  "linked_request_id": "req_xxx",
  "linked_contractor_id": "...",
  "received_by": "usr_cre",
  "received_at": "...",
  "metadata": {...}
}

```

## `cashbook`

```

{
  "expense_id": "exp_xxx",
  "project_id": "proj_xxx",
  "project_name": "anbu",
  "category": "labour | material | petty_cash | income",
  "amount": 1000,
  "approved_amount": 1000,
  "payment_method": "cash | cheque | bank",
  "cheque_no": "...",
  "bank_ref": "...",
  "contractor_id": "...",
  "stage_id": "stg_xxx",
  "stage_name": "Foundation",
  "is_multi_stage_bill": false,
  "rab_group_id": "grp_xxx",
  "linked_request_ids": ["req_xxx"],
  "stage_breakdown": [{"stage_id": "...", "stage_name": "...", "amount": 1000, "request_id": "..."}],
  "status": "approved",
  "source": "wo_stage_release",
  "payment_entries": [{"method": "cheque", "amount": 1000, "bank_ref": "", "cheque_ids": ["chq_xxx"]}],
  "payment_date": "...",
  "created_at": "..."
}

```

## `payment\_stages` (CRE client-side ladder)

```

{
  "stage_id": "pst_xxx",
  "project_id": "proj_xxx",
  "stage_no": 1,
  "stage_name": "Booking Advance",
  "scheduled_amount": 250000,
  "due_date": "...",
  "linked_section_id": null,
  "status": "requested | received | approved | rejected",
  "payment_record": {...},
  "source": "manual | additional"
}

```

## `notifications`

```
{
  "notification_id": "ntf_xxx",
  "user_id": "usr_xxx",
  "type": "rab_pm_approved",
  "title": "RAB-02 awaiting QC",
  "body": "...",
  "link": "/projects/proj_xxx",
  "is_read": false,
  "created_at": "..."
}
```

## 4. Authentication Header

Every authenticated request sends:

Authorization: Bearer <JWT>

JWT body: {user\_id, role, name, email, exp}.

## 5. Response Envelopes

Successful responses return the payload directly (no **data** envelope).

Errors return:

```
{ "detail": "Human-readable error message" }
```

HTTP codes used: 200 (OK) · 400 (validation) · 403 (forbidden by role) · 404 (not found) · 409 (state conflict).

## 6. Pagination & Filtering Conventions

- Most list endpoints accept **limit** (default 100-500) + optional filters as query params.
- Date filters use ISO strings (**start\_date=2026-01-01&end\_date=2026-02-01**).
- Status filters accept comma-separated lists where supported.

## 7. ID Conventions

| Entity          | Prefix  |
|-----------------|---------|
| User            | `usr_`  |
| Project         | `proj_` |
| Work Order      | `wo_`   |
| Stage           | `stg_`  |
| Payment Request | `req_`  |
| Section         | `sec_`  |
| Cheque          | `chq_`  |
| Cashbook        | `exp_`  |
| RAB Group       | `grp_`  |
| Notification    | `ntf_`  |

All IDs are 12-character hex (**uuid4().hex[:12]**).

\_End of API reference.\_